



Causal Analysis and Resolution (CAR)

CAR Overview

Required PA Information

Intent

Identifies causes of selected outcomes and takes action to either prevent recurrence of undesirable outcomes or ensure recurrence of positive outcomes.

Value

Addresses causes of issues, eliminating rework and directly improving quality and productivity.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

CAR 1.1 Identify and address causes of selected outcomes.

Level 2

CAR 2.1 Select outcomes for analysis.

CAR 2.2 Analyze and address causes of outcomes.

Level 3

CAR 3.1 Determine causes of selected outcomes by following an organizational process.

CAR 3.2 Propose actions to address identified causes.

CAR 3.3 Implement selected action proposals.

CAR 3.4 Record causal analysis and resolution data.

CAR 3.5 Submit improvement proposals for changes proven to be effective.

Level 4

CAR 4.1 Perform root cause analysis of selected outcomes using statistical and other quantitative techniques.

CAR 4.2 Evaluate the effect of implemented actions on process performance using statistical and other quantitative techniques.

Level 5

- CAR 5.1 Use statistical and other quantitative techniques to evaluate other solutions and processes to determine if the resolution should be applied to optimize performance across the organization.

Additional PA Explanatory Information

Causal analysis and resolution activities include:

- Determining causes of selected outcomes
- Proposing actions to address identified causes
- Implementing selected action proposals
- Recording causal analysis and resolution data
- Submitting improvement proposals for changes proven to be effective

It is more cost-effective to prevent defects and problems from occurring than to detect defects and problems after they have been introduced.

Since it is impractical to perform causal analysis on all outcomes, select targets by analyzing tradeoffs between estimated investments and returns. Causal analysis and resolution activities provide a mechanism for projects and the organization to evaluate their processes and look for improvements that can be implemented. When improvements are judged to be effective, following the processes that guide them, the information is submitted to the organizational level for potential deployment in the organizational processes.

Related Practice Areas

Managing Performance and Measurement (MPM)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Agile teams identify and address blockers and collect Sprint data during the iteration and review it in the retrospective. A typical agile retrospective reviews what went wrong, what worked well, and selects improvements for the next Sprint.

For example, a team that is consistently unable to complete the work defined for each Sprint would look for various causes, such as chronic distractions, using unreliable velocity data, poorly defined user stories, exceeding team capacity, or underestimating complexity. These underlying causes would be identified, analyzed, ranked, and addressed.

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams typically conduct reviews at the end of each iteration based on the framework or methodology chosen, such as Scrum, Kanban, SaFe, etc. For example, in Scrum, retrospectives occur at the end of each Sprint. During these retrospectives, teams identify progress and work impediments or blockers to resolve these during the next Sprint. The challenge that DevSecOps teams face when performing causal analysis is the perception that when during a retrospective someone says, "Let's get to the root of the problem." This often means: Who is to blame? This "blame game" is counter to a positive DevSecOps culture. The target of root cause analysis should be focused on the process and not the people.

The challenge with root cause analysis in modern software development is it may not consistently address complexity of the software and so often does not provide solutions quickly enough. In a DevSecOps environment where iterations are short, and Continuous Integration / Continuous Delivery (CI/CD) pipelines, micro services, containers, etc., are in constant flux, getting to the root cause is more challenging due to the limitations of data at specific points in time, and due to the complex interactions between all system components, interfaces or connections, applications, and environments.

DevSecOps teams typically address causal analysis incrementally, focusing on what is actionable and can be corrected in the next Sprint. For example, failures in gate-check points in the CI/CD pipeline could be potential outcomes to initiate analysis, and simple techniques like Five Whys can be used. Some issues may require multiple Sprints to correct or require collaboration across multiple DevSecOps teams. For example: the inability to identify errors in the early stages; lack of team member collaboration, lack of ability to identify vulnerabilities, etc.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Causal analysis and resolution activities may include evaluating acquirer processes that interface or connect with supplier processes. When acquirers and suppliers jointly perform causal analysis, it can lead to improvement actions such as:

- Supplier improving its processes to conduct the project more effectively
- Acquirer improving its supplier interfaces or connections
- Changes in jointly used tools or technologies

Level 1

CAR 1.1

Required Practice Information

Practice Statement

Identify and address causes of selected outcomes.

Value

Increases likelihood of achieving business objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Outcomes can be either positive or negative. Significant differences from what was expected indicate where the project can improve its performance by:

- Determining why it went well and how to change to incorporate the experience into the normal behavior
- Determining why expectations were not met and the changes needed to meet them

Example Activities

Example Activities	Further Explanation
Select outcomes that differed from expectation.	
Investigate causes of outcomes.	
Address causes and record changes made to address the causes.	

Example Work Products

Example Work Products	Further Explanation
List of investigated outcomes	May include: • Outcomes • Causes • Changes made

Level 2

CAR 2.1

Required Practice Information

Practice Statement

Select outcomes for analysis.

Value

Focuses efforts on the outcomes with the greatest impact on achieving objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

This activity can be triggered by an event or planned periodically, such as at the beginning of a new phase, iteration, or task.

Examples of when to perform causal analysis include:

- During the task when problems or successes warrant a causal analysis
- When a work product significantly deviates from its requirements
- When more defects than anticipated escape from earlier phases
- When process performance exceeds expectations
- When a process does not meet its Quality and Process Performance Objectives (QPPOs)

Example Activities

Example Activities	Further Explanation
Define the scope of the analysis.	The scope may include: • Definition of issue or success • Affected stakeholders • Affected target
Collect relevant data.	
Determine which outcomes to analyze further.	When determining which outcomes to analyze further, consider their: • Source • Impact • Frequency of occurrence • Similarity • Cost of analysis • Time and resources needed

Example Activities	Further Explanation
	• Safety considerations • Security considerations • Performance Examples of methods for selecting outcomes include: • Pareto analysis • Histograms • Box and whisker plots for attributes • Failure modes and effects analysis (FMEA) • Design of experiment • Cause and effect analysis

Example Work Products

Example Work Products	Further Explanation
Analysis results	The results of high-level analysis selected for more detailed causal analysis.
Outcomes selected for further analysis	

Context Specific

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

Retrospectives are time-boxed activities and because of this, once all issues from the previous Sprint have been identified, teams typically need to use previously defined criteria to prioritize and pick their top two to three items for further analysis. Use of criteria is beneficial in helping uncover and address root causes that can develop into a larger issue in upcoming Sprints and to prioritize corrective action in upcoming Sprints. It may be several Sprints before the team addresses all key issues. It also may take several Sprints before the team can determine if their corrective actions are effective.

CAR 2.2

Required Practice Information

Practice Statement

Analyze and address causes of outcomes.

Value

Reduces cost and time to meet objectives more efficiently.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Causal analysis of selected issues is best performed shortly after the problem is first identified, while the event is still recent enough to be carefully investigated.

The formality of and effort required for causal analysis can vary greatly and can be determined by factors including:

- Stakeholders who perform the work
- Risks
- Complexity
- Frequency
- Availability of data
- Available resources

Example Activities

Example Activities	Further Explanation
Identify the affected stakeholders and involve them.	Causal analysis is typically performed by those who have the best understanding of the selected outcome and who are responsible for performing the task.
Perform causal analysis.	
Identify and analyze potential issues or successes.	
Implement selected actions.	
Assess the impact of the actions on performance.	
Communicate results.	

Example Work Products

Example Work Products	Further Explanation
List of affected stakeholders	
Identified causes	These are the results of analyzing potential issues and outcomes.
Actions to take	Action lists may include potential: • Cost and schedule impacts • Performance impacts

Level 3

CAR 3.1

Required Practice Information

Practice Statement

Determine causes of selected outcomes by following an organizational process.

Value

Increases likelihood of meeting objectives by promoting successes and avoiding problems.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

A cause is a fundamental reason for the occurrence of a problem or success. Causal analysis can include qualitative and basic measurement data.

Example Activities

Example Activities	Further Explanation
Identify and involve stakeholders.	
Collect data.	
Follow an organizational process to perform causal analysis.	Consider looking at individual outcomes as well as grouping the outcomes. Negative outcomes can be influenced by: • Inadequate training and skills • Breakdown of communication • Not accounting for all details of a task • Making mistakes in manual procedures, e.g., keyboard entry • Process deficiency • Inadequate resource allocation • Incomplete, ambiguous, or unclear contractual requirements • Ineffective management of changes to the supplier agreement Positive outcomes can be influenced by: • New approaches to projects • Process automation • System or tool upgrades

Example Activities	Further Explanation
	• Pilots • Process improvements • Performance improvements Examples of methods to determine causes include: • Cause-and-effect (fishbone) diagrams • Check sheets • 5 Whys If possible, based on the scope, look at the outcomes in several ways to ensure all potential causes are investigated. Where appropriate, look for patterns of causes across functions.
Record causes.	

Example Work Products

Example Work Products	Further Explanation
List of causes	Include selected outcomes and analysis results.
List of affected stakeholders	

CAR 3.2

Required Practice Information

Practice Statement

Propose actions to address identified causes.

Value

Reduces cost and time by preventing negative outcomes or producing positive outcomes.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Based on the analysis results, develop action proposals that will address selected outcomes in accordance with an organizational process. These action proposals address the removal of causes so that they do not recur.

Example Activities

Example Activities	Further Explanation
Develop an action proposal.	Action proposals may include: • Identified processes • Training • Tools • Methods • Solutions Common ways to address underlying causes include: • Changing a process to remove error-prone steps • Updating a process based on previous successes • Deploying the results of successful pilots • Eliminating non-value-added tasks • Automating all or part of a process • Reordering process activities • Adding process steps, such as task kickoff meetings, to review common problems and actions to prevent them
Record action proposals.	

Example Work Products

Example Work Products	Further Explanation
Action proposals	May include: • Originator of the action proposal • Affected stakeholders • Description of necessary tasks • Description of the outcome to be addressed • Description of the cause • Causes and actions categories • Phase identified • Description of the action • Time, cost, and other resources required to implement the action proposal • Expected benefits from implementing the action proposal • Estimated cost of not fixing the problem or leveraging the success

CAR 3.3

Required Practice Information

Practice Statement

Implement selected action proposals.

Value

Implements changes that have the most impact on increasing the likelihood of meeting objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

These actions should prevent or reduce the occurrence of negative outcomes or increase the occurrence of positive outcomes.

Example Activities

Example Activities	Further Explanation
Analyze action proposals and determine their priorities.	Criteria for prioritizing action proposals include: • Implications of not addressing the outcome • Cost to implement actions to address the outcome • Risk implications • Expected impact on quality
Select action proposals to be implemented.	Include criteria to be used for deciding which action proposals to implement.
Develop action plans for implementing the selected action proposals.	Action plans may include: • People responsible for implementation • Detailed description of the action • Description of necessary tasks • Description of the affected areas • Affected stakeholders • Schedule • Cost expected • Estimated cost of not addressing the cause • Description of implementation actions • Expected impact on performance • Identified needed pilots
Implement action plans.	To implement action plans: • Update the processes • Review the results

Example Activities	Further Explanation
	Track action items to closure Actions may be assigned to members of the causal analysis team, members of the project team, or other members of the organization.
Look for similar causes that may exist in other processes and solutions and take action as appropriate.	

Example Work Products

Example Work Products	Further Explanation
Action proposals selected for implementation	
Action plans	
Updated process assets	

Related Practice Areas

Decision Analysis and Resolution (DAR)

Process Asset Development (PAD)

CAR 3.4

Required Practice Information

Practice Statement

Record causal analysis and resolution data.

Value

Records and communicates improvement efforts across the organization, leveraging savings and increasing productivity.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Collect data to know that you are:

- Improving project performance
- Preventing selected problems from recurring

- Leveraging superior performance
- Providing sufficient context for future use

Example Activities

Example Activities	Further Explanation
Record causal analysis data and make the data available for use.	Involve the use of qualitative and basic measurement data analysis with the causal analysis.

Example Work Products

Example Work Products	Further Explanation
Cause analysis and resolution records	May include: • Data on outcomes that were analyzed • Rationale for decisions • Action proposals • Action plans resulting from action proposals • Cost of analysis and resolution activities • Measures of changes to the process performance of the recorded process

CAR 3.5

Required Practice Information

Practice Statement

Submit improvement proposals for changes proven to be effective.

Value

Reduces disruptions and rework and allows projects to learn from each other and increase productivity.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Ensure that other projects in the organization can benefit from the improvements.

Example Activities

Example Activities	Further Explanation
Submit improvement proposals	May include: • Areas that were analyzed including their context

Example Activities	Further Explanation
	• Solution selection decisions made including the context of their solution • Actions selected including trade-off contexts evaluated • Monitoring tasks including unintended side effects • Results achieved including performance information with context, even if expected outcome is not fully achieved by the solution

Example Work Products

Example Work Products	Further Explanation
Improvement proposals	

Related Practice Areas

Process Management (PCM)

Level 4

CAR 4.1

Required Practice Information

Practice Statement

Perform root cause analysis of selected outcomes using statistical and other quantitative techniques.

Value

Improves the likelihood that the project will meet its quality and process performance objectives.

Additional Required Information

Perform root cause analysis using data and results from statistical and quantitative analysis techniques to evaluate, select, and implement action plans and measure results. This should be done in addition to basic measurement and qualitative analysis activities. Address outcomes including deficiencies in process stability and capability, deficiencies in performance relative to objectives, and unexpected positive results.

Explanatory Practice Information

Additional Explanatory Information

Root cause analysis typically depends on the availability of data, baselines, and models that can be used in the analysis. Actions to take can range significantly in terms of effort and time needed to determine, plan, and implement. It is difficult to know how much time is needed without an initial analysis of the deficiencies.

Example Activities

Example Activities	Further Explanation
Perform root cause analysis.	To understand the impact on stability and capability and to determine the reasons for positive and negative outcomes. Process performance baselines and models are used in: • Diagnosing deficiencies • Diagnosing positive outcomes • Identifying possible solutions • Predicting future work and process performance • Evaluating potential actions
Identify and analyze potential actions.	
Identify measures of effectiveness.	

Example Activities	Further Explanation
Implement selected actions.	Update solutions or processes.

Example Work Products

Example Work Products	Further Explanation
Process and project performance analyses	Should include: • Statistical and quantitative analysis • Data visualization used to understand process and project performance and trends
Identified root causes	
Identified measures of effectiveness	May include: • Impact on performance • Impact on meeting Quality and Process Performance Objectives (QPPOs) • Impact on stability and capability
Action plans	Include changes to solutions or processes.
Updated solutions or processes	

Related Practice Area

Managing Performance and Measurement (MPM)

CAR 4.2

Required Practice Information

Practice Statement

Evaluate the effect of implemented actions on process performance using statistical and other quantitative techniques.

Value

Maximizes the likelihood of meeting quality and process performance objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Evaluate the effect of changes to verify that the process change is statistically significant.

Example Activities

Example Activities	Further Explanation
Measure and analyze the change in process performance of the project's effected processes.	Determines if the selected change has positively influenced process performance. Statistical and other quantitative techniques, e.g., hypothesis testing can be used to compare the before and after baselines to assess the statistical significance of the change.
Determine the impacts of the change on achieving the project's Quality and Process Performance Objectives (QPPOs).	Determines whether the selected change has positively influenced the ability of the project to meet its QPPOs. Process performance models can aid in the evaluation through prediction of impacts and return on investment.
Submit process improvement proposals for the organization when the implemented actions are effective.	

Example Work Products

Example Work Products	Further Explanation
Analysis of change in process performance	
Organizational improvement proposals	

Related Practice Area

Managing Performance and Measurement (MPM)

Level 5

CAR 5.1

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to evaluate other solutions and processes to determine if the resolution should be applied to optimize performance across the organization.

Value

Leverages improvements across the organization to minimize cost and risk.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Learn from root cause analysis and determine if resolutions from historical projects and solutions can be applied to other projects, processes, and solutions in the organization.

Example Activities

Example Activities	Further Explanation
Identify similar processes or solutions.	
Analyze to determine candidates for change and prioritize them.	
Apply changes to selected processes or solutions and communicate results.	

Example Work Products

Example Work Products	Further Explanation
Identified candidate processes and solutions	
Results of changes	

